THEIVANAI AMMAL COLLEGE FOR WOMEN

(An Autonomous Institution, Affiliated to Annamalai University, Chidambaram - Tamil Nadu)

(Accredited with 'A+' Grade (4th Cycle) by NAAC, Recognized under 2(f) and 12(B) by UGC, Approved by AICTE, New Delhi)

Villupuram-Tamil Nadu

DEPARTMENT OF COMPUTER SCIENCE

This is to certify that Ms. …**PRADEEPA. S**……,Reg. No: …**20524UCSC028**…….., a student of III B.Sc. Computer Science, has successfully completed the practice-based practical learning component of **Advanced Database Management System – Practical (UCSR513)** during the academic year 2026–2027.

The practical examination will be held on **16.10.2026**

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1. DDL COMMANDS

Table Creation:

SQL > create table flight_booking (cid number(10), cname varchar(20), ctp number(10));

```
Table created.
```

0.10 seconds

Table Description:

SQL> desc flight_booking;

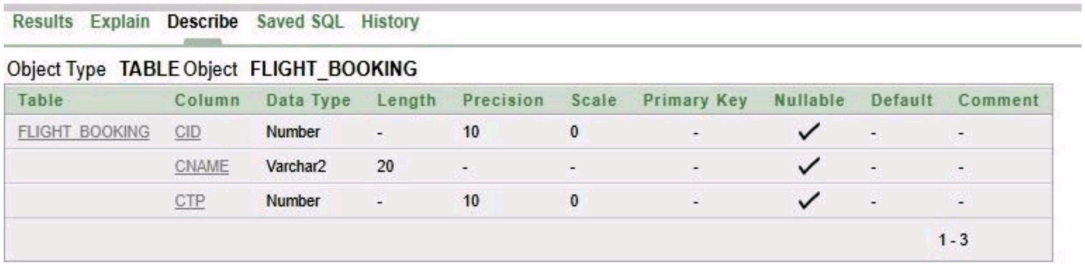


Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FLIGHT_BOOKING	CID	Number	-	10	0	-	✓	-	-
	CNAME	Varchar2	20	-	-	-	✓	-	-
	CTP	Number	-	10	0	-	✓	-	-

1 - 3

Table Alteration:

Method 1(Add)

SQL> alter table flight_booking add cphn number(10);

```
Table altered.
```

0.05 seconds

SQL> select*from flight_booking;

Results Explain Describe Saved SQL History

CID	CNAME	CTP	CPHN
111	Savitha	30000	-
112	Priya	40000	-
113	Raji	20000	-
114	Bavana	50000	-

4 rows returned in 0.05 seconds

Method 2(Drop)

SQL> alter table student drop column class;

Results Explain Describe Saved SQL History

```
Table dropped.
```

0.27 seconds

SQL> select*from flight_booking;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
111	Savitha	30000
112	Priya	40000
113	Raji	20000
114	Bavana	50000

4 rows returned in 0.01 seconds

Rename Table:

SQL> rename flight_booking to f_b;

Results Explain Describe Saved SQL History

```
Statement processed.
```

0.05 seconds

Truncate Table:

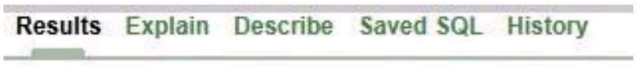
SQL> truncate table f_b;

```
Table truncated.
```

0.26 seconds

Drop Table:

SQL> drop table f_b;




```
Table dropped.
```

0.36 seconds

2. DML COMMANDS

Table Insertion:

SQL> insert into flight_booking values(111, 'Savitha', 30000);

1 row(s) inserted.

SQL> insert into flight_booking values(112, 'Priya', 40000);

1 row(s) inserted.

SQL> insert into flight_booking values(113, 'Raji', 20000);

1 row(s) inserted.

SQL> insert into flight_booking values(114, 'Bavana', 50000);

1 row(s) inserted.

Table updating:

SQL> update f_b set cname='Varsha' where cid=111;

Results Explain Describe Saved SQL History

1 row(s) updated.

0.01 seconds

SQL> select*from f_b;

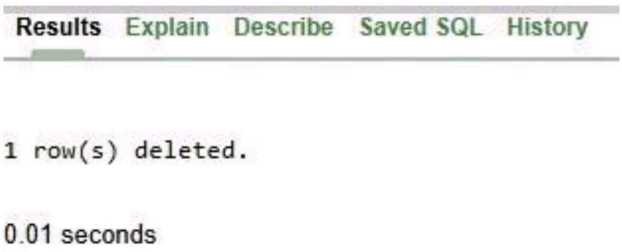
Results Explain Describe Saved SQL History

CID	CNAME	CTP
111	Varsha	30000
112	Priya	40000
113	Raji	20000
114	Bavana	50000

4 rows returned in 0.00 seconds

Table Deletion:

SQL>delete from f_b where cid=111;



3. DQL COMMAND

Table View:

SQL> select*from f_b;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
112	Priya	40000
113	Raji	20000
114	Bavana	50000

3 rows returned in 0.00 seconds

4. TCL COMMAND

ROLLBACK

SQL> delete from f_b where cid=111;

Results Explain Describe Saved SQL History

```
1 row(s) deleted.
```

0.00 seconds

SQL> select*from f_b;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
112	Priya	40000
113	Raji	20000
114	Bavana	50000

3 rows returned in 0.00 seconds

SQL> rollback;

Results Explain Describe Saved SQL History

```
Statement processed.
```

0.00 seconds

SQL> select*from f_b;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
111	Savitha	30000
112	Priya	40000
113	Raji	20000
114	Bavana	50000

4 rows returned in 0.00 seconds

SAVEPOINT

SQL> begin

savepoint p;

end;

Results Explain Describe Saved SQL History

```
Statement processed.
```

0.00 seconds

SQL> insert into f_b values(115,'Gaja',80000);

Results Explain Describe Saved SQL History

```
1 row(s) inserted.
```

0.00 seconds

SQL> select*from f_b;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
111	Savitha	30000
112	Priya	40000
113	Raji	20000
114	Bavana	50000
115	Gaja	80000

5 rows returned in 0.00 seconds

SQL> rollback to p;

SQL> select*from f_b;

Results Explain Describe Saved SQL History

CID	CNAME	CTP
111	Savitha	30000
112	Priya	40000
113	Raji	20000
114	Bavana	50000

4 rows returned in 0.00 seconds

COMMIT

SQL> commit;

Results Explain Describe Saved SQL History

```
Statement processed.
```

0.00 seconds

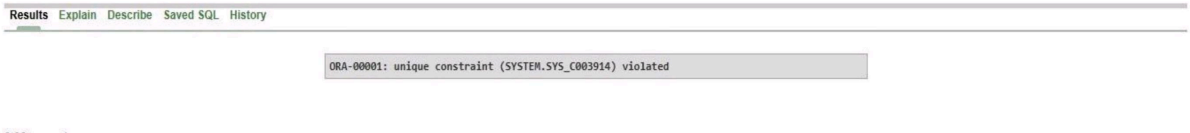
5. CONSTRAINTS

UNIQUE:

SQL> create table food(fid number(10), fname varchar(20) unique);

SQL> insert into food values(01,'dosa');

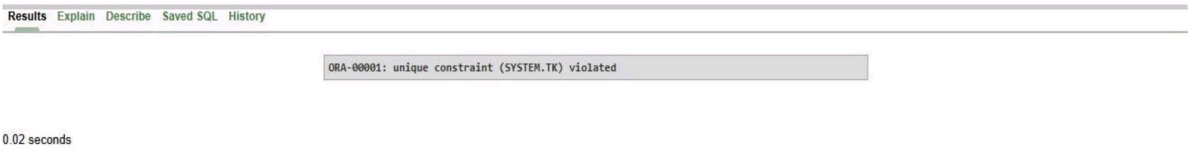
SQL> insert into food values(02,'dosa');



PRIMARY KEY:

SQL> alter table food add constraint tk primary key(fid);

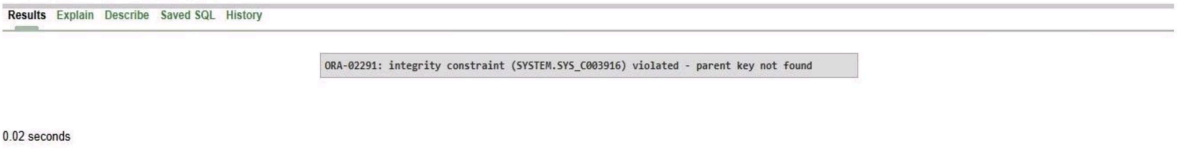
SQL> insert into food values(01,'puri');



FOREIGN KEY:

SQL> create table hotel(fid number(10), fname varchar(20), foreign key(fid) references food(fid));

SQL> insert into hotel values(03,'pongal');



CHECK:

SQL> alter table food add constraint ck check(fid<20);

SQL> insert into food values(21,'idly');

6. JOINS

TABLE - 1

SQL> create table employee(eno number(10), empname varchar(20), city varchar(20));

Table Created

TABLE INSERTION

SQL> insert into employee values(1001, 'Savitha', 'Pondy')

1 row(s) inserted.

SQL> insert into employee values(1002, 'Priya', 'Villupuram')

1 row(s) inserted.

SQL> insert into employee values(1003, 'Raji', 'Madukkarai')

1 row(s) inserted.

VIEW TABLE

SQL> select * from employee

Results Explain Describe Saved SQL History

ENO	EMPNAME	CITY
1001	Savitha	Pondy
1002	Priya	Villupuram
1003	Raji	Madukkarai

3 rows returned in 0.02 seconds CSV Export

TABLE - 2

SQL> create table emp_sal(emp_name varchar(20),dept varchar(20),salary number(10))

TABLE INSERTION

SQL> insert into emp_sal values('Savitha','CS',60000)

1 row(s) inserted.

SQL> insert into emp_sal values('Bavana','IT',70000)

1 row(s) inserted.

SQL> insert into emp_sal values('Raji','BCA',80000)

1 row(s) inserted.

VIEW TABLE

SQL> select * from emp_sal

Results Explain Describe Saved SQL History

EMP_NAME	DEPT	SALARY
Saviha	CS	60000
Bavana	IT	70000
Raji	BCA	80000

3 rows returned in 0.20 seconds CSV Export

INNER JOIN

SQL> select * from employee inner join emp_sal on employee.empname = emp_sal.emp_name;

OUTPUT

Results Explain Describe Saved SQL History

ENO	EMPNAME	CITY	EMP_NAME	DEPT	SALARY
1001	Savitha	Pondy	Savitha	CS	60000
1003	Raji	Madukkarai	Raji	BCA	80000

2 rows returned in 0.00 seconds CSV Export

LEFT OUTER JOIN

SQL> select e.empname, s.salary from employee e left outer join emp_sal s on employee.empname=emp_sal.emp_name;

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME	SALARY
Savitha	60000
Raji	80000
Priya	-

3 rows returned in 0.00 seconds CSV Export

RIGHT OUTER JOIN

SQL> select e.empname, e.city, s.salary from employee e right outer join emp_sal s on e.empname=s.emp_name;

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME	CITY	SALARY
Savitha	Pondy	60000
Raji	Madukkarai	80000
-	-	70000

3 rows returned in 0.00 seconds CSV Export

NATURAL FULL JOIN

SQL> select * from employee natural full join emp_sal;

OUTPUT

Results Explain Describe Saved SQL History

ENO	EMPNAME	CITY	EMP_NAME	DEPT	SALARY
1001	Savitha	Pondy	Savitha	CS	60000
1001	Savitha	Pondy	Bavana	IT	70000
1001	Savitha	Pondy	Raji	BCA	80000
1002	Priya	Villupuram	Savitha	CS	60000
1002	Priya	Villupuram	Bavana	IT	70000
1002	Priya	Villupuram	Raji	BCA	80000
1003	Raji	Madukkarai	Savitha	CS	60000
1003	Raji	Madukkarai	Bavana	IT	70000
1003	Raji	Madukkarai	Raji	BCA	80000

9 rows returned in 0.04 seconds CSV Export

7. AGGREGATE FUNCTIONS

SUM

SQL> select sum(salary) "Total" from emp_sal;

OUTPUT

Results Explain Describe Saved SQL History

Total
210000

1 rows returned in 0.70 seconds CSV Export

MAX

SQL> select max (salary)"Maximum Salary" from emp_sal;

OUTPUT

Results Explain Describe Saved SQL History

Maximum Salary
80000

1 rows returned in 0.00 seconds CSV Export

MIN

SQL> select min (salary)"Minimum Salary" from emp_sal;

OUTPUT

Results Explain Describe Saved SQL History

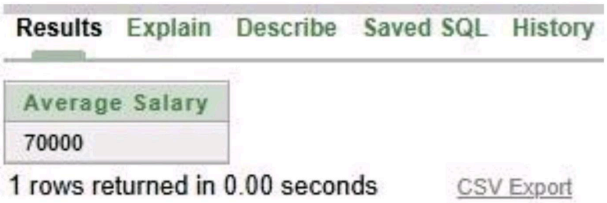
Minimum Salary
60000

1 rows returned in 0.00 seconds CSV Export

AVERAGE

SQL> select avg (salary)"Average Salary" from emp_sal;

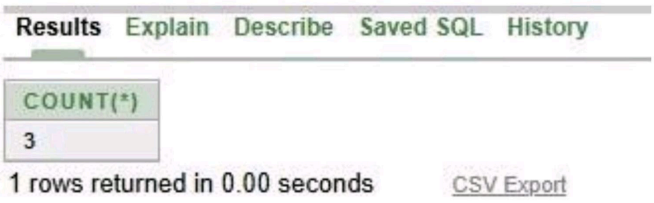
OUTPUT



COUNT

SQL> select count(*) from emp_sal;

OUTPUT



8. VIEW CREATIONS

SQL> create view employee_view as select eno, empname from employee;

OUTPUT

Results Explain Describe Saved SQL History

```
View created.
```

0.07 seconds

SQL> select * from employee_view;

OUTPUT

Results Explain Describe Saved SQL History

ENO	EMPNAME
1001	Savitha
1002	Priya
1003	Raji

3 rows returned in 0.02 seconds CSV Export

9. SET OPERATORS

UNION

SQL> select * from employee UNION select * from emp_sal

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME	CITY
Bavana	IT
Priya	Villupuram
Raji	BCA
Raji	Madukkarai
Savitha	CS
Savitha	Pondy

6 rows returned in 0.00 seconds CSV Export

UNION ALL

SQL> select * from employee UNION ALL select * from emp_sal

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME	CITY
Bavana	IT
Priya	Villupuram
Raji	BCA
Raji	Madukkarai
Savitha	CS
Savitha	Pondy

6 rows returned in 0.00 seconds CSV Export

INTERSECT

SQL> select empname from employee INTERSECT select emp_name from emp_sal

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME
Raji
Savitha

2 rows returned in 0.02 seconds CSV Export

MINUS

SQL> select * from employee MINUS select * from emp_sal

OUTPUT

Results Explain Describe Saved SQL History

EMPNAME	CITY
Priya	Villupuram
Raji	Madukkarai
Savitha	Pondy

3 rows returned in 0.02 seconds CSV Export

10. STRING FUNCTION

REPLACE

SQL> select replace('hello world', 'world', 'oracle') from dual;

OUTPUT

Results Explain Describe Saved SQL History

REPLACE('HELLOWORLD','WORLD','ORACLE')
hello oracle

1 rows returned in 0.14 seconds CSV Export

SUBSTR

SQL> select substr('computer', 1, 4) from dual;

OUTPUT

Results Explain Describe Saved SQL History

SUBSTR('COMPUTER',1,4)
comp

1 rows returned in 0.00 seconds CSV Export

INITCAP

SQL> select initcap('hello world') from dual;

OUTPUT

Results Explain Describe Saved SQL History

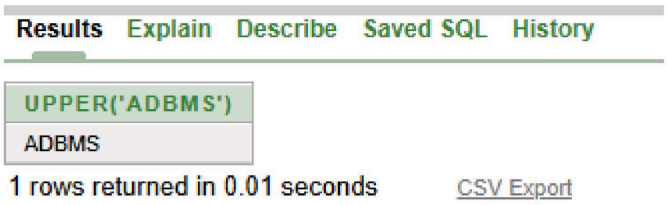
INITCAP('HELLOWORLD')
Hello World

1 rows returned in 0.01 seconds CSV Export

UPPER

SQL> select upper('adbms') from dual;

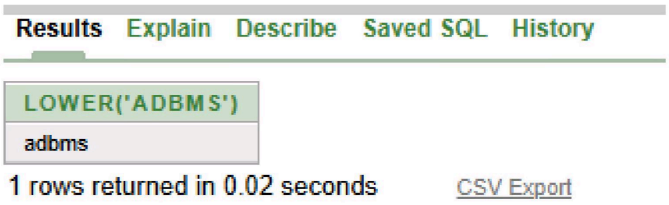
OUTPUT



LOWER

SQL> select lower('ADBMS') from dual;

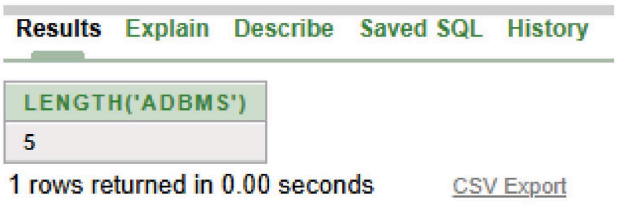
OUTPUT



LENGTH

SQL> select length('ADBMS') from dual;

OUTPUT

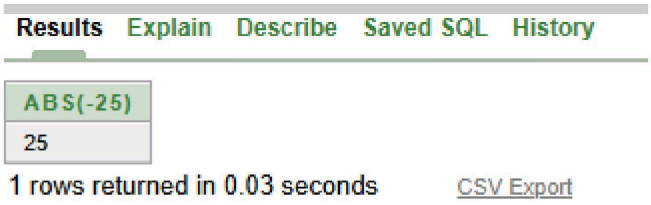


11. NUMERIC FUNCTION

ABSOLUTE

SQL> select abs(-25) from dual;

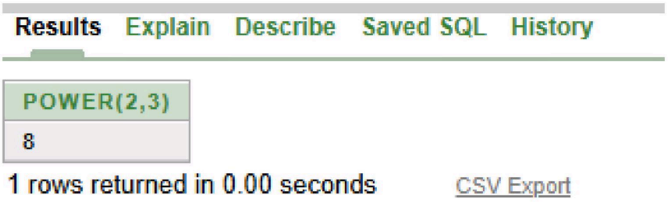
OUTPUT



POWER

SQL> select power(2, 3) from dual;

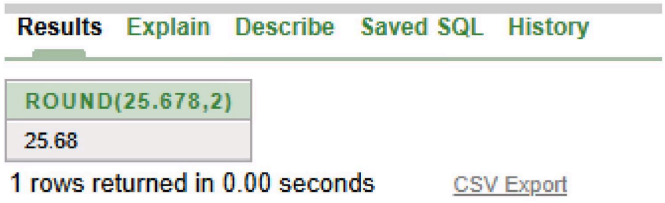
OUTPUT



ROUND

SQL> select round(25.678, 2) from dual;

OUTPUT



12. NESTED SUBQUERY

TABLE 1:

```
SQL> create table student_sub (student_id number, student_name varchar2(30), dept_id
     number, mark number);
SQL> insert into student_sub values (101, 'savitha', 1, 85);
SQL> insert into student_sub values (102, 'priya', 1, 90);
SQL> insert into student_sub values (103, 'gaja', 2, 75);
SQL> insert into student_sub values (104, 'bavana', 2, 80);
SQL> insert into student_sub values (105, 'raji', 3, 95);
```

TABLE 2:

```
SQL> create table department_sub (dept_id number, dept_name varchar2(30));
SQL> insert into department_sub values (1, 'computer science');
SQL> insert into department_sub values (2, 'information technology');
SQL> insert into department_sub values (3, 'artificial intelligence');
```

IN

```
SQL> select student_name, mark from student_sub where dept_id in (select dept_id from
     department_sub where dept_name = 'computer science');
```

OUTPUT

Results Explain Describe Saved SQL History

STUDENT_NAME	MARK
priya	90
savitha	85

2 rows returned in 0.10 seconds CSV Export

WHERE

SQL> select student_name, mark from student_sub where mark > (select avg(mark) from student_sub);

OUTPUT

Results Explain Describe Saved SQL History

STUDENT_NAME	MARK
priya	90
raji	95

2 rows returned in 0.00 seconds CSV Export

GROUP BY

SQL> select dept_id, count(*) as total_students from student_sub where dept_id in (select dept_id from department_sub) group by dept_id;

OUTPUT

Results Explain Describe Saved SQL History

DEPT_ID	TOTAL_STUDENTS
1	2
2	2
3	1

3 rows returned in 0.07 seconds CSV Export

HAVING

SQL> select dept_id, avg(mark) as average_mark from student_sub group by dept_id having avg(mark) > (select avg(mark)from student_sub);

OUTPUT

Results Explain Describe Saved SQL History

DEPT_ID	AVERAGE_MARK
1	87.5
3	95

2 rows returned in 0.01 seconds CSV Export

ORDER BY

SQL> select student_name, mark from student_sub where mark > (select avg(mark) from student_sub) order by mark desc;

OUTPUT

Results Explain Describe Saved SQL History

STUDENT_NAME	MARK
raji	95
priya	90

2 rows returned in 0.00 seconds CSV Export

BETWEEN

SQL> select student_name, mark from student_sub where mark between (select min(mark)from student_sub) and (select avg(mark) from student_sub);

OUTPUT

Results Explain Describe Saved SQL History

STUDENT_NAME	MARK
savitha	85
gaja	75
bavana	80

3 rows returned in 0.02 seconds CSV Export

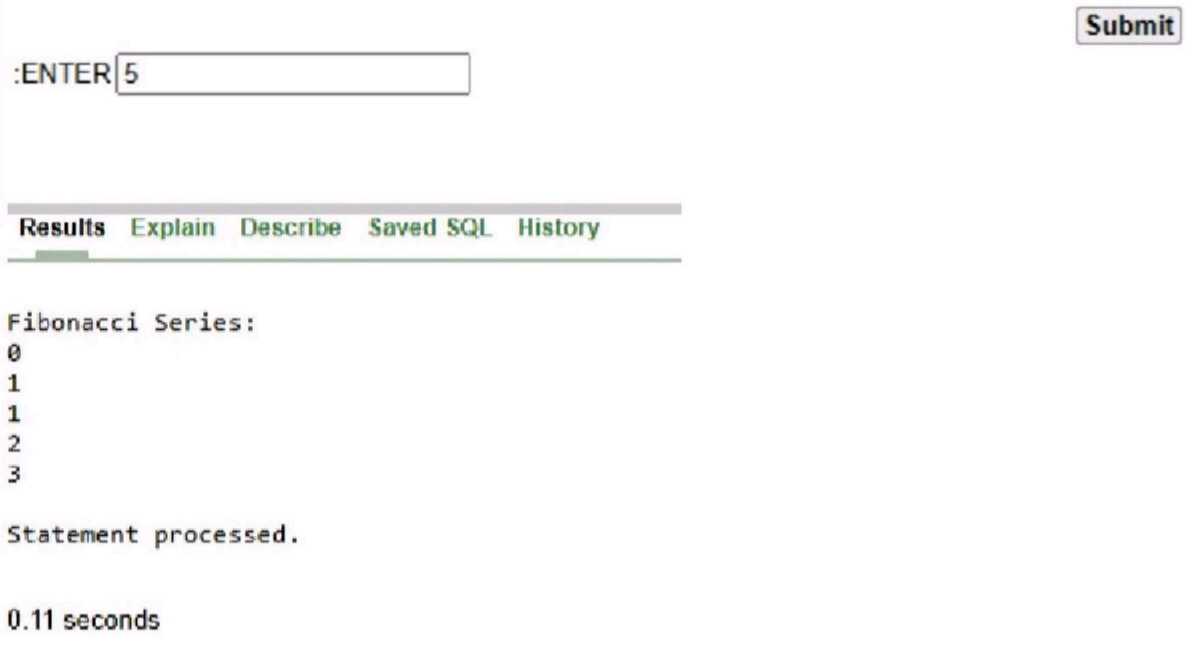
PL/SQL

1. FIBONACCI SERIES

PROGRAM

```
declare
   n number := :Enter;
   a number := 0;
   b number := 1;
   c number;
begin
   dbms_output.put_line('fibonacci series:');
   for i in 1..n loop
      dbms_output.put_line(a);
      c := a + b;
      a := b;
      b := c;
   end loop;
end;
```

OUTPUT



2. FACTORIAL

PROGRAM

```
declare
   n number := :Enter;
   fact number := 1;
begin
   for i in 1..n loop
      fact := fact * i;
   end loop;
   dbms_output.put_line('factorial = ' || fact);
end;
```

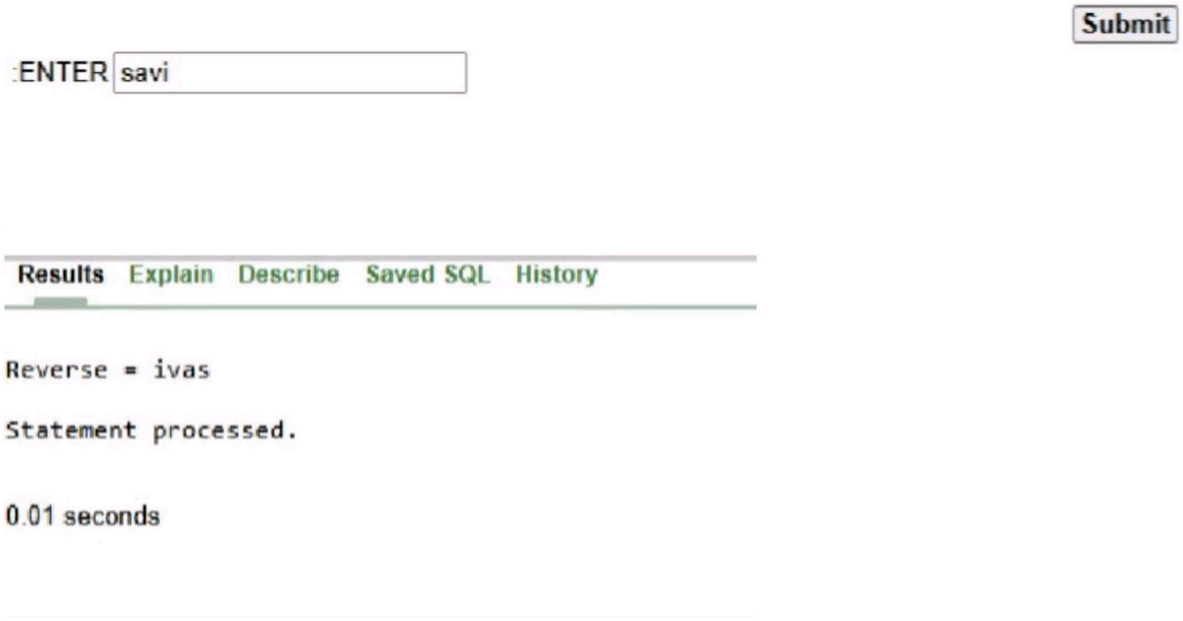
OUTPUT

3. STRING REVERSE

PROGRAM

```
declare
   str varchar(10) := :Enter;
   rev varchar(10) := ' ';
begin
   for i in reverse 1..length(str) loop
      rev := rev || substr(str, i, 1);
   end loop;
   dbms_output.put_line('reverse = ' || rev);
end;
```

OUTPUT

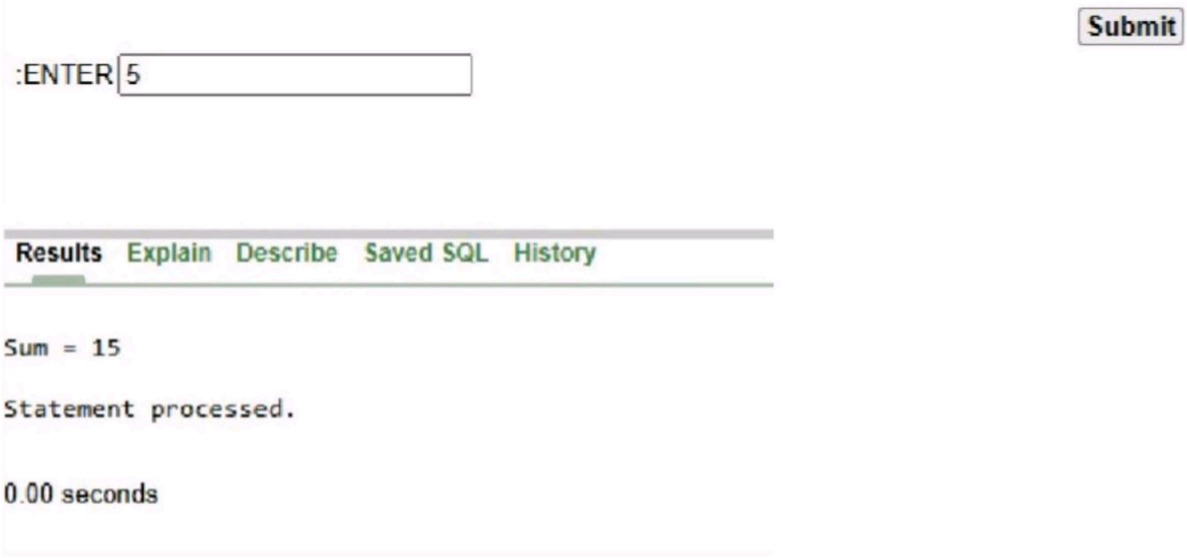


4. SUM OF SERIES

PROGRAM

```
declare
   n number := :Enter;
   total number := 0;
begin
   for i in 1..n loop
      total := total + i;
   end loop;
   dbms_output.put_line('sum = ' || sum);
end;
```

OUTPUT



5. TRIGGER

TABLE CREATION

SQL> create table voter(vid number(10), vname varchar(20), vcity varchar(20));

Table Created

TABLE INSERTION

SQL> insert into voter values(1001, 'Savitha', 'pondy');

1 row(s) inserted

SQL> insert into voter values(1002,'Varsha','Villupuram');

1 row(s) inserted

SQL> insert into voter values(1003,'Jessi','Pondi');

1 row(s) inserted

TABLE VIEW

Select * from voter

Results Explain Describe Saved SQL History

VID	VNAME	VCITY
1001	Savitha	pondy
1002	Varsha	Villupuram
1003	Jessi	Pondi

3 rows returned in 0.05 seconds CSV Export

TRIGGER CREATION

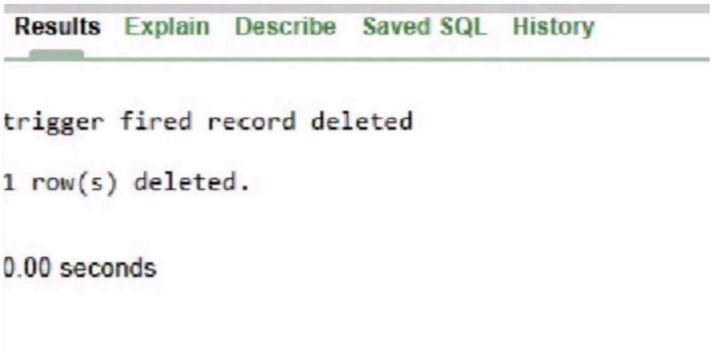
```
create or replace trigger tud
after update or delete on voter
declare
 msg varchar(20):='trigger fired';
begin
 if updating then
dbms_output.put_line(msg||'record updated');
elsif deleting then
dbms_output.put_line(msg||'record deleted');
end if;
end tud;
```

UPDATE RECORD

SQL> update voter set vcity='Pondi' where vid=1002

DELETION RECORD

SQL> delete from voter where vid=1003;



6. STUDENT MARK AND GRADE CALCULATION

PROGRAM

```
declare
   mark number := :Enter_Mark;
   grade varchar(10);
begin
   if mark >= 90 then
      grade := 'a';
   elsif mark >= 80 then
      grade := 'b';
   elsif mark >= 70 then
      grade := 'c';
   elsif mark >= 60 then
      grade := 'd';
   elsif mark >= 50 then
      grade := 'e';
   else
      grade := 'f';
   end if;
   dbms_output.put_line('mark = ' || mark);
   dbms_output.put_line('grade = ' || grade);
end;
```

OUTPUT

Results Explain Describe Saved SQL History

```
mark = 85
grade = b

Statement processed.
```

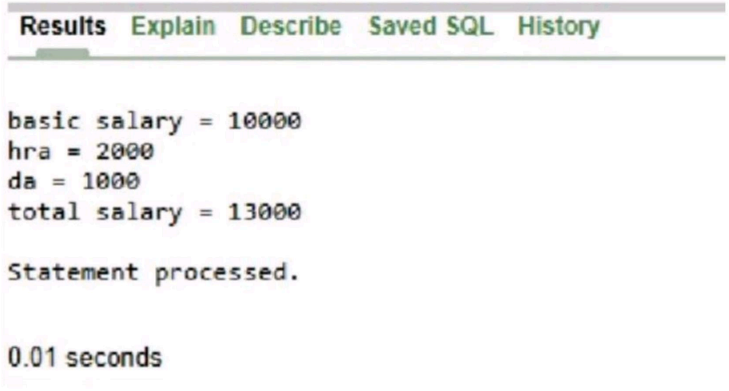
0.02 seconds

7. EMPLOYEE SALARY CALCULATION

PROGRAM

```
declare
   basic number := :Enter_Basic;
   hra number;
   da number;
   total number;
begin
   hra := basic * 0.20;
   da := basic * 0.10;
   total := basic + hra + da;
   dbms_output.put_line('basic salary = ' || basic);
   dbms_output.put_line('hra = ' || hra);
   dbms_output.put_line('da = ' || da);
   dbms_output.put_line('total salary = ' || total);
end;
```

OUTPUT

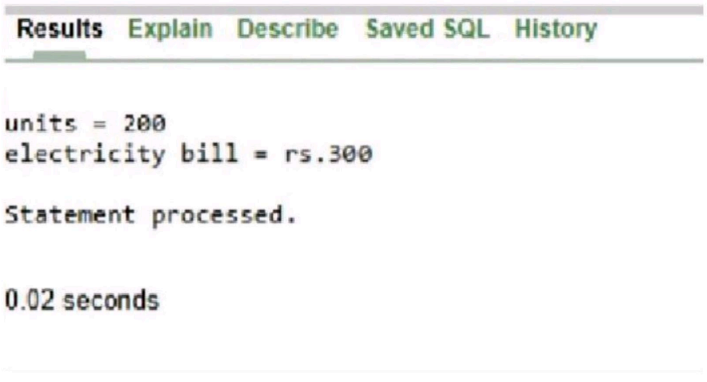


8. ELECTRICITY BILL CALCULATION

PROGRAM

```
declare
   units number := :Enter_Units;
   bill number;
begin
   if units <= 100 then
      bill := units * 1;
   elsif units <= 200 then
      bill := 100 + (units - 100) * 2;
   elsif units <= 500 then
      bill := 300 + (units - 200) * 3;
   else
      bill := 1200 + (units - 500) * 5;
   end if;
   dbms_output.put_line('units = ' || units);
   dbms_output.put_line('electricity bill = rs.' || bill);
end;
```

OUTPUT

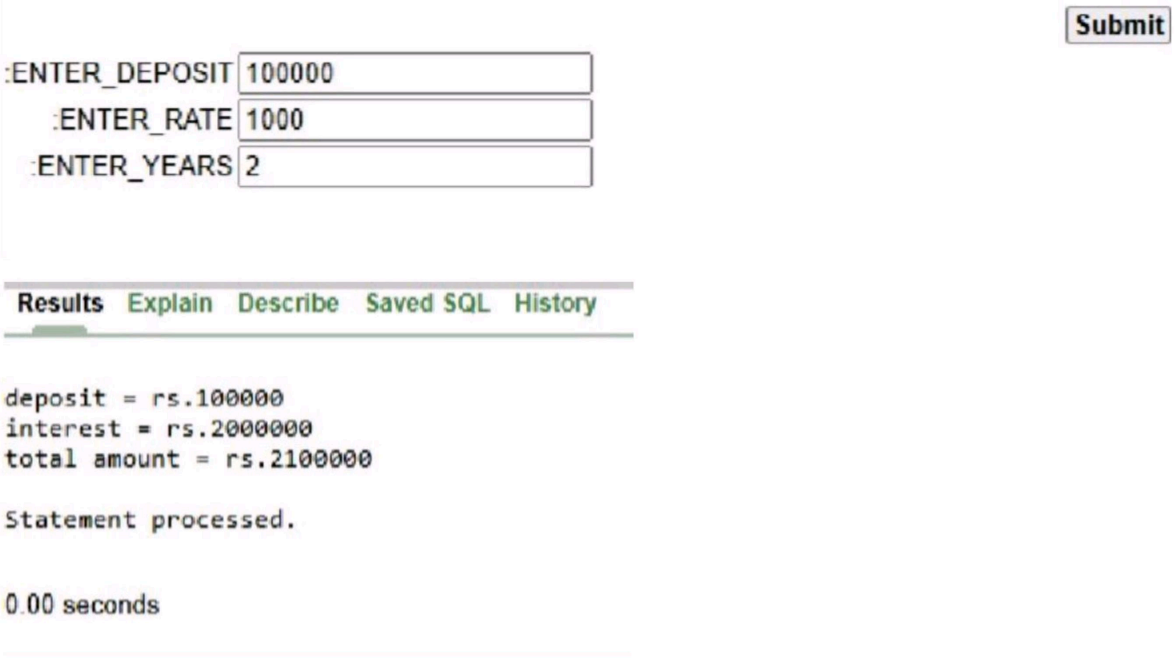


9. BANK DEPOSIT AND INTEREST CALCULATION

PROGRAM

```
declare
   deposit number := :Enter_Deposit;
   rate number := :Enter_Rate;
   years number := :Enter_Years;
   interest number;
   amount number;
begin
   interest := (deposit * rate * years) / 100;
   amount := deposit + interest;
   dbms_output.put_line('deposit = rs.' || deposit);
   dbms_output.put_line('interest = rs.' || interest);
   dbms_output.put_line('total amount = rs.' || amount);
end;
```

OUTPUT

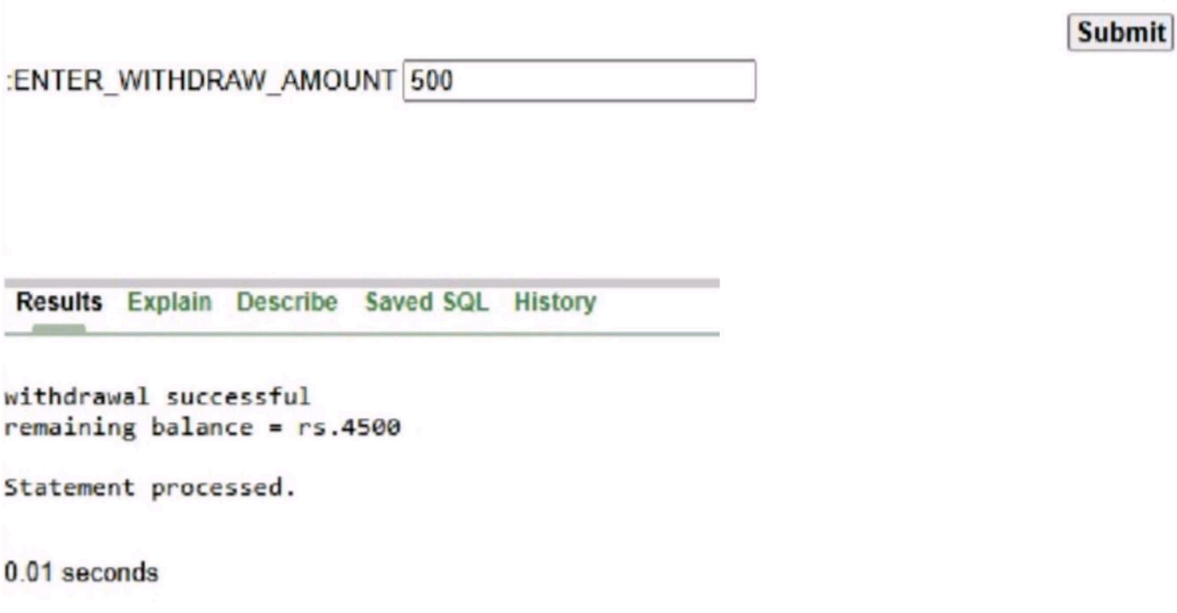


10. ATM WITHDRAWAL

PROGRAM

```
declare
   balance number := 5000;
   withdraw number := :Enter_Withdraw;
begin
   if withdraw <= 0 then
      dbms_output.put_line('invalid amount');
   elsif withdraw > balance then
      dbms_output.put_line('insufficient balance');
   else
      balance := balance - withdraw;
      dbms_output.put_line('withdrawal successful');
      dbms_output.put_line('remaining balance = rs.' || balance);
   end if;
end;
```

OUTPUT



11. PRODUCT DISCOUNT CALCULATION

PROGRAM

```
declare
   price number := :Enter_Price;
   discount number;
   final_price number;
begin
   if price >= 5000 then
      discount := price * 0.20;
   elsif price >= 3000 then
      discount := price * 0.15;
   elsif price >= 1000 then
      discount := price * 0.10;
   else
      discount := 0;
   end if;
   final_price := price - discount;
   dbms_output.put_line('price = rs.' || price);
   dbms_output.put_line('discount = rs.' || discount);
   dbms_output.put_line('final price = rs.' || final_price);
end;
```

OUTPUT

Results Explain Describe Saved SQL History

```
price = rs.5000
discount = rs.1000
final price = rs.4000

Statement processed.
```

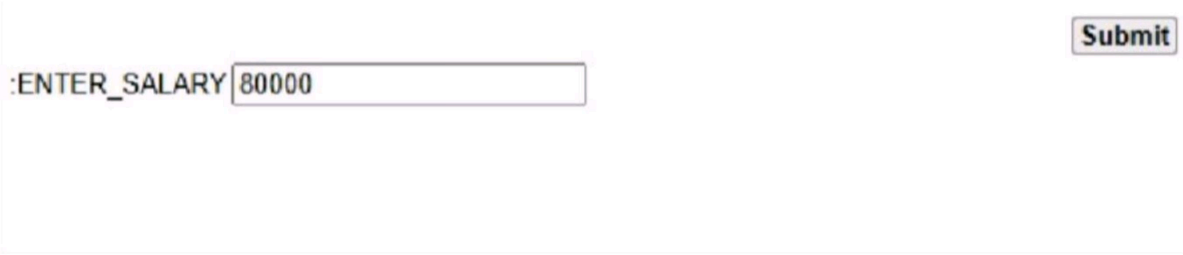
0.02 seconds

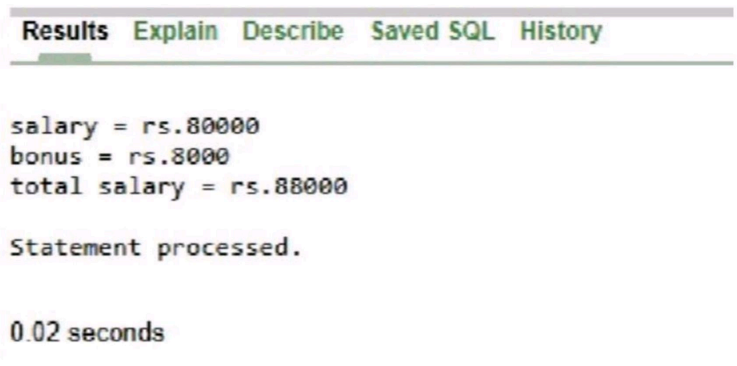
12. EMPLOYEE BONUS CALCULATION

PROGRAM

```
declare
   salary number := :Enter_Salary;
   bonus number;
   total number;
begin
   if salary >= 50000 then
      bonus := salary * 0.10;
   else
      bonus := salary * 0.05;
   end if;
   total := salary + bonus;
   dbms_output.put_line('salary = rs.' || salary);
   dbms_output.put_line('bonus = rs.' || bonus);
   dbms_output.put_line('total salary = rs.' || total);
end;
```

OUTPUT



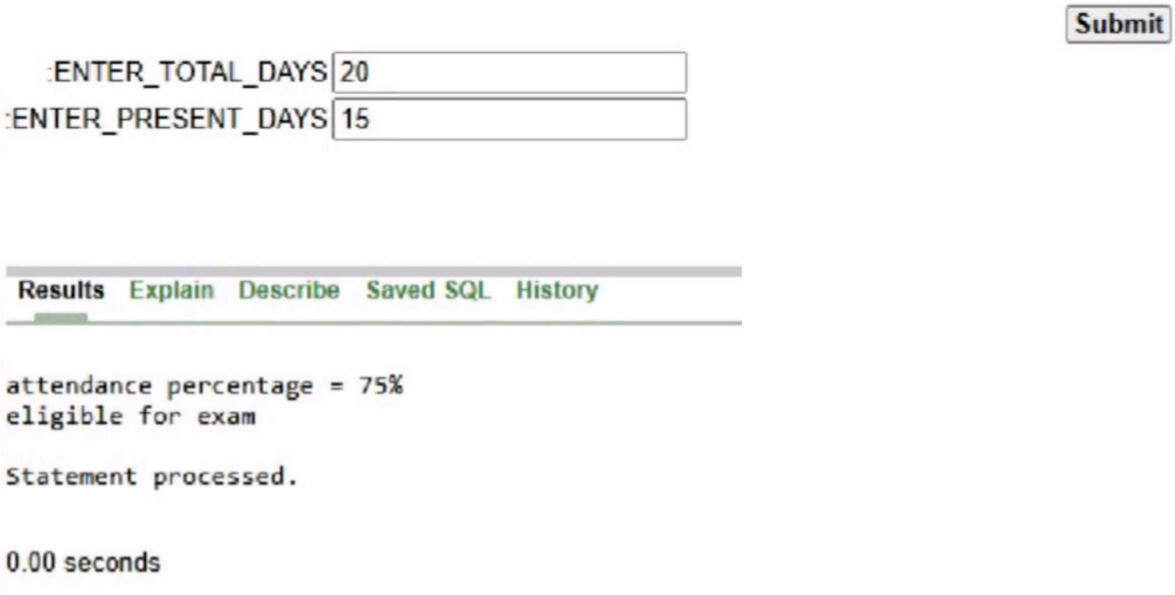


13. STUDENT ATTENDANCE ELIGIBILITY

PROGRAM

```
declare
   total_days number := :Enter_Total_days;
   present_days number := :Enter_Present_days;
   percentage number;
begin
   percentage := (present_days / total_days) * 100;
   dbms_output.put_line('attendance percentage = ' || percentage || '%');
   if percentage >= 75 then
      dbms_output.put_line('eligible for exam');
   else
      dbms_output.put_line('not eligible for exam');
   end if;
end;
```

OUTPUT



14. ARMSTRONG NUMBER

PROGRAM

```
declare
   n number := :ENTER;
   temp number;
   rem number;
   total number := 0;
begin
   temp := n;

   while temp > 0 loop
      rem := mod(temp, 10);
      total := total + rem * rem * rem;
      temp := trunc(temp / 10);
   end loop;

   if total = n then
      dbms_output.put_line('armstrong number');
   else
      dbms_output.put_line('not an armstrong number');
   end if;
end;
```

OUTPUT

Results Explain Describe Saved SQL History

```
armstrong number

Statement processed.
```

0.06 seconds

15. CURSOR

TABLE CREATION

SQL> create table customer (id number, name varchar2(20), address varchar2(30), salary number);

TABLE INSERTION

SQL> insert into customer values (1, 'savitha', 'pondy', 20000);

SQL> insert into customer values (2, 'priya', 'villupuram', 25000);

SQL> insert into customer values (3, 'gaja', 'chennai', 30000);

SQL> select * from customer;

Results Explain Describe Saved SQL History

ID	NAME	ADDRESS	SALARY
1	savitha	pondy	20000
2	priya	villupuram	25000
3	gaja	chennai	30000

3 rows returned in 0.08 seconds CSV Export

IMPLICIT CURSOR

```
declare
   total_rows number;
begin
   update customer
   set salary = salary + 500;

   if sql%notfound then
      dbms_output.put_line('no customers selected');
   else
      total_rows := sql%rowcount;
      dbms_output.put_line(total_rows || ' customers selected');
   end if;
end;
```

OUTPUT

```
3 customers selected

1 row(s) updated.

0.07 seconds
```

SQL> select * from customer;

ID	NAME	ADDRESS	SALARY
1	savitha	pondy	20500
2	priya	villupuram	25500
3	gaja	chennai	30500

3 rows returned in 0.00 seconds CSV Export

EXPLICIT CURSOR

```
declare
   c_id customer.id%type;
   c_name customer.name%type;
   c_addr customer.address%type;

   cursor c_customers is
      select id, name, address from customer;
begin
   open c_customers;

   loop
      fetch c_customers into c_id, c_name, c_addr;

      exit when c_customers%notfound;

      dbms_output.put_line(c_id || ' ' || c_name || ' ' || c_addr);
```

```
    end loop;

    close c_customers;
end;
```

OUTPUT

```
1 savitha pondy
2 priya villupuram
3 gaja chennai

Statement processed.


0.03 seconds
```

16. CHECK EVEN OR ODD NUMBER

PROGRAM

```
declare
   n number := :ENTER;
begin
   if mod(n, 2) = 0 then
      dbms_output.put_line('Even Number');
   else
      dbms_output.put_line('Odd Number');
   end if;
end;
```

OUTPUT

17. PALINDROME CHECKER

PROGRAM

```
declare
   n number := :ENTER;
   temp number;
   rev number := 0;
   rem number;
begin
   temp := n;

   while temp > 0 loop
      rem := mod(temp, 10);
      rev := rev * 10 + rem;
      temp := trunc(temp / 10);
   end loop;

   if n = rev then
      dbms_output.put_line('Palindrome');
   else
      dbms_output.put_line('Not Palindrome');
   end if;
end;
```

OUTPUT

Results Explain Describe Saved SQL History

```
Palindrome

Statement processed.
```

0.02 seconds

18. PRIME NUMBER

PROGRAM

```
declare
   n number :=:ENTER;
   count number := 0;
begin
   for i in 1..n loop
      if mod(n, i) = 0 then
         count := count + 1;
      end if;
   end loop;

   if count = 2 then
      dbms_output.put_line('Prime Number');
   else
      dbms_output.put_line('Not a Prime Number');
   end if;
end;
```

OUTPUT

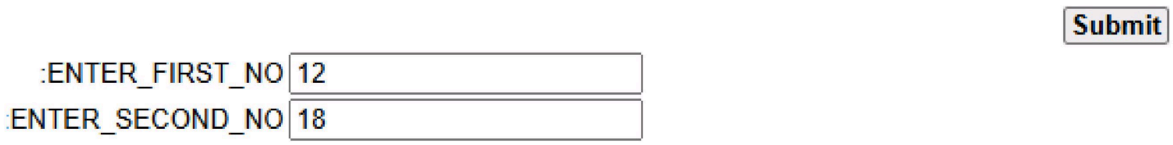
19. GCD

PROGRAM

```
declare
   a number := :ENTER_FIRST_NO;
   b number := :ENTER_SECOND_NO;
   temp number;
begin
   while b != 0 loop
      temp := mod(a, b);
      a := b;
      b := temp;
   end loop;

   dbms_output.put_line('GCD = ' || a);
end;
```

OUTPUT



```
GCD = 6

Statement processed.

0.00 seconds
```

20. OPERATORS

PROGRAM

```
declare
   a number := :ENTER_FIRST_NO;
   b number := :ENTER_SECOND_NO;
   choice number := :ENTER_CHOICE_NO;
begin
   if choice = 1 then
      dbms_output.put_line('addition = ' || (a + b));

   elsif choice = 2 then
      dbms_output.put_line('subtraction = ' || (a - b));

   elsif choice = 3 then
      dbms_output.put_line('multiplication = ' || (a * b));

   elsif choice = 4 then
      dbms_output.put_line('division = ' || (a / b));

   elsif choice = 5 then
      dbms_output.put_line('modulus = ' || mod(a, b));

   elsif choice = 6 then
      if a > b then
         dbms_output.put_line('a is greater');
      else
         dbms_output.put_line('b is greater or equal');
      end if;

   elsif choice = 7 then
      if a = b then
```

```
        dbms_output.put_line('both are equal');
    else
        dbms_output.put_line('both are not equal');
    end if;

  else
    dbms_output.put_line('invalid choice');
  end if;
end;
```

OUTPUT